# SHG Mode Survey: Field Guide

shg_mode_survey · PDF record of the claude.ai field-guide artifact, 2026-10-12

Surveys every pump->SH mode pair for a given waveguide cross-section: finds every phase-matched pair in the SH window, then ranks by efficiency and loss. Two outcome families: guided->guided (the SH mode is itself bound, scored by NCE) and radiative (Cerenkov/leaky -- flagged and quantified, not discarded).

### Wavelength convention

SH wavelength sweeps the target window (default 215-235nm); pump = 2x SH wavelength (430-470nm). Mode labels TMmn/TEmn (m=lateral nodes, n=vertical). AlN's c-axis is y, so d33 couples Ey.Ey -> Py.

#### STATUS AT A GLANCE

| | |
|---|---|
| **ridge** | done, Step 1/1b; strongest family under calibrated SH loss |
| **loaded** | TiO2 strip ruled out (SH-absorbing); SiN ready, not run |
| **hybrid** | geometry fixed 2026-10-10 (now genuinely embedded); re-run pending |
| **pedestal** | legacy stacked structure (the old hybrid bug); kept for reference only |
| **materials** | AlN's own measured SH absorption explored, REJECTED; 100 dB/cm calibrated value used instead |

#### WHAT THIS REPORT COVERS, AND WHAT IT DOESN'T

Covers: status, loss model (incl. the AlN rejection/calibration story), geometry families with real cross-sections, and the gotchas most likely to bite again. Deliberately omits operational material (how to launch a run, notebook vs. hands-off, terminal commands) -- that lives on the live artifact, where it's actually actionable; see PLAN.md or the artifact link for those. Figures of merit (NCE, peak efficiency) are defined once, canonically, in input_assumptions_report.pdf section 0 -- not repeated here.

# Where things stand

**ridge**

Done for Step 1/1b. Best design: TM00->TM04, h=350/w=500nm, 227.9nm, NCE 4.8 %/W/cm^2. Still the strongest family once AlN's own SH-band loss is included: median calibrated peak efficiency ~41,000x above `loaded`'s, best-to-best ~4,700x (aln_loss_report.pdf). No TM02/TM20 partner phase-matches near the window.

**loaded**

Geometry verified against EMode's own render. TiO2 strip sweep (8 geometries): lossless NCE far exceeds ridge (159 %/W/cm^2) but TiO2's own bulk SH absorption is so strong `loaded` stays worst-in-class even under the calibrated 100 dB/cm assumption. Next: SiN strip (ready, not run) or ScAlN (blocked on d33(Sc)).

**hybrid**

AlN core embedded on the sides+above by a SiN shell -- tests the Honda et al. 2025 TM00->TM20 mechanism. A geometry bug (core stacked ON the shell, not embedded) went undetected through several runs; fixed 2026-10-10. Not re-run since the fix -- queued, Simulation Study Plan row 6, launching with the calibrated SH-loss default.

**pedestal**

The stacked structure `hybrid` was accidentally building pre-fix -- renamed to its own family 2026-10-12 rather than discarded (real hardware, wrong mechanism). 3 runs, 416 refined crossings, AlN-lossless only: field exports absent on both machines, re-solve costed at ~9.5hr EMode time for a family being superseded.

**materials**

AlN/sapphire/SiO2 built in. TiO2, ScAlN (not wired into a geometry yet) fit from literature. AlN's own UV k: AM187-LT6 MOCVD sample fit and wired as an opt-in mechanism, but its SH-band implication (~100,000 dB/cm) was checked against real device output and found 3-4 orders of magnitude too high. NOT USED -- a flat, empirically-calibrated 100 dB/cm placeholder is the standing assumption instead.

**EMode env.**

Desktop on v1.0.5 as of 2026-10-12 (updated from v1.0.0, independently, since the last check -- version drift between machines is a recurring silent risk; nothing tracks it). Two EMode defects remain open, not re-tested against v1.0.5.

# Loss model

Three base mechanisms, summed per mode per band: EMode-native roughness scattering (em.scattering(), Volume Current Method); post-processed interface absorption (dislocations, C/Si/O impurities, pump-band only); confinement-weighted bulk material absorption (a lossy region's own k(lambda) x the mode's power fraction inside it, a post-solve perturbation -- re-solving EMode with a lossy material crashes em.FDM() outright on every version tried, and the one mechanism that doesn't crash (shape(loss_dB_per_m=...), deprecated) was found to corrupt SH-mode re-identification when the loss is large enough to shift the eigenvalue search -- TiO2's case dropped a nominally-unchanged crossing's overlap_shape 0.051->0.0036 with a different mode label.

## AlN's own SH-band absorption: measured, rejected, replaced (2026-10-12)

AlN was assumed lossless in every result before this. A real MOCVD sample (AM187-LT6, team's own ellipsometry_data repo, lowest non-negative k_e among all MOCVD positions) gave a measured k_e, fit to 3 Lorentzians, wired the same way as the bulk mechanism above. Implies ~100,000 dB/cm of SH-band loss for the best-matching real device geometry -- checked against that device's actual measured SH output (~200nW best case, ~10s of nW typically, ~200mW pump, 10-50% unmeasured coupling) by inverting shg_physics.loss_limited_length at the real fixed device length. Bounds true SH loss to ~15-680 dB/cm depending on coupling -- 3-4 orders of magnitude below AM187's number.

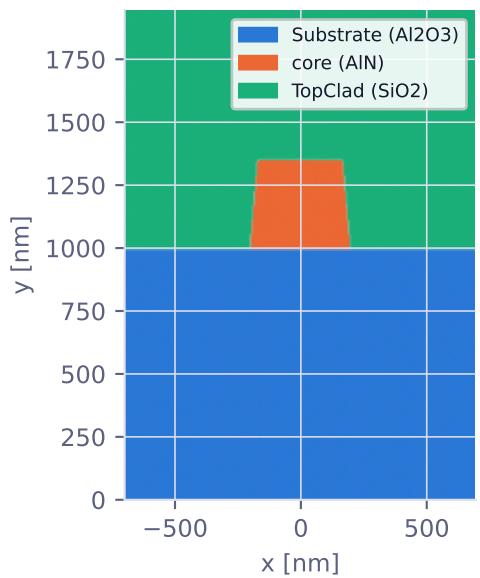
### Working value: 100 dB/cm, confinement-weighted, SH-band only

A round number inside the empirically-bounded range, not a measurement of any specific mechanism -- geometry.sh_absorption_calibrated_dB_per_m, confinement-weighted like the bulk mechanism above, added on top of (not replacing) native scattering. The standing assumption for every new run from here on. Full derivation: input_assumptions_report.pdf section 3b; PLAN.md's 2026-10-12 "empirical SH-loss bound" entries.

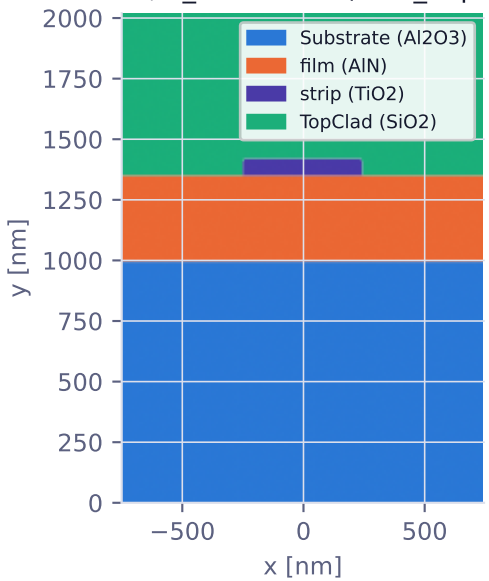
# Geometry families

family name + a params dict; every family implements the same build()/region_masks()/chi2_map()/bbox() interface

Ridge (h=350, w=400 nm)

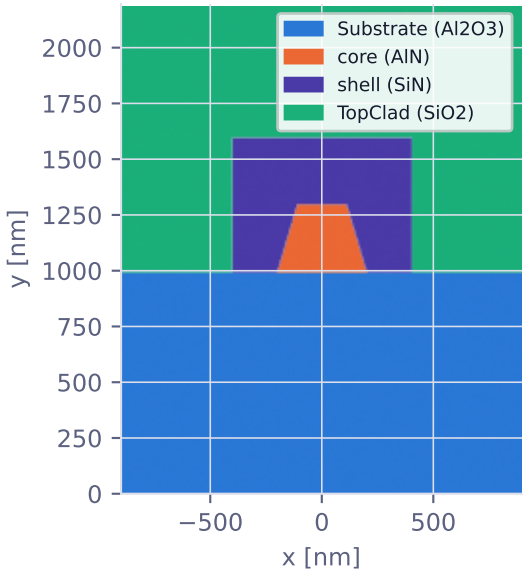



Loaded, h_A=full film (etch_depth=0)

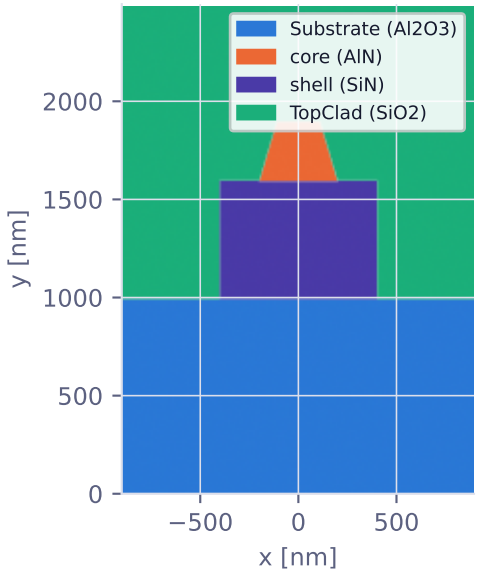



ridge: bare AlN core on substrate, no A/B layers. loaded: AlN film capped by a narrower loading strip (TiO2 shown; SH-absorbing, ruled out). hybrid: core wrapped on the sides AND above by the shell (corrected 2026-10-10). pedestal: the same params, but core sits ON TOP of the shell -- what hybrid built before the fix.

hybrid, corrected embedded (h=300, w=400)




pedestal, legacy stacked (same params)



# Gotchas most likely to bite again

## A geometry param can auto-stack instead of embed

em.shape()'s position defaults to (0,'auto'), which STACKS a new shape on top of whatever's there; an explicit position is CENTER-referenced, not bottom-referenced. This is exactly how `hybrid` silently built the stacked `pedestal` structure for months -- it ran, gave plausible numbers, looked fine until someone checked the actual rendered cross-section. Verify any new embedding geometry two independent ways (em.get_shape() metadata + an em.confinement() power-fraction check).

## EMode version matters, and nothing tracks it

Check (Get-Item "...\EMode.exe").VersionInfo before debugging an EMode-API-shaped issue as code, on EITHER machine, every time -- a prior session's note can already be stale (found 2026-10-12: desktop had moved to v1.0.5 without it being logged).

## Binary field exports do not travel with git, or scale through chat tools

runs/*/exports/*.npz is gitignored (~270MB per 8-geometry sweep). A cloud-drive MCP tool's per-file download doesn't scale past a handful of files (each lands as base64 text in context). What works: download the whole folder as a zip from the web UI, copy over the local filesystem directly.

## A great NCE can be a simulation-window artifact

Box modes (quantized by the window's hard walls, not real confinement) can carry a genuine local field enhancement, so NCE/overlap_shape alone don't catch them -- check shg_physics.lateral_edge_ratio; survey.py/summarize_run.py now filter automatically, but a pre-2026-10-05 run's crossings weren't checked at all.

## Raw mode index is not a stable identifier

Can shift by 7+ from a modest width or wavelength change. Every tracking step identifies modes by field-overlap similarity against the previous step, never by raw index -- new analysis code against the exports should do the same.

# Results index

**input_assumptions_report.pdf**
Loss model, material dispersion, figures of merit. Canonical -- every other report should cite it.

**aln_loss_report.pdf**
Lossless vs. calibrated-SH-loss comparison, across every reprocessed run.

**simulation_study_plan.pdf**
The 10-row geometry x material design matrix, sequencing, standing assumptions.

**field_guide_report.pdf**
This document.

**runs/<name>/summary/summary.md**
Per-run ranking tables, the human-readable version of that run's CSVs.

**runs/<name>/crossings.csv**
Every phase match: NCE, overlap, losses, L_opt, peak efficiency, per refined/screened row.

**PLAN.md**
The dated lab notebook -- every finding above traces back to an entry there. "Open items" is the live backlog.

Live artifact (interactive, operational instructions included):
`claude.ai/artifact/AcW4tCTXUH3xXTPhq5zi4a -- updated alongside this PDF, 2026-10-12.`